NUMERICAL METHODS - MIDTERM EXAM

Q1: Find the root of $f(x) = x^2 \cos(2x) + 1$ using the Bisection Method…

Using the ***Bisection Method***..

(b) In Market it te the Taylor serics

using the first first appeenee veries

Note work in 5 siguificunt figures

$$\ln(\cos(2x)) = -2x^2 - \frac{4}{3}x^4 - \frac{64}{45}x^6 - \frac{256}{315}x^8 - \dots$$

(Find the abseloteonl truncation errers and

True Value · *Approximate Valoe*

Absolute Error % - *Trubeotion Error %*

$f(x) = x^2 \cos(2x) + 1$

$x_L = 1.1$

$x_R = 1.5$

$x_4 = 5$

$f(x_1) = (1.1)^2 \cos(2(1.1)) + 1 = 0.58385$

$f(x_2) = (1.25)^2 \cos(2(1.25)) + 1 = -0.35179$

$f(x_1)\ f(x_2) < 0 \rightarrow x_R = x_2$, $x_4 = x_4$

$x_L = 1.1$

$x_R = \frac{1.1 + 1.25}{2} = 1.175$

$x_3 = 1.25$

$f(x_L)$ 0.58385

$f(x_3) = (1.125)^2 \cos(2(1.125)) + 1 = 0.20494$

$f(x_L)\ f(x_3) > 0 \rightarrow x_L = x_3$, $x_4 = x_4$

$x_L = 1.125$

$x_R = \frac{1.125 + 1.25}{2} = 1.1875$

$x_4 = 1.25$ ← R3

Q2- Solve the following systems of linear equations (12 Marks)

(a) (6 Marks): Use the ***Cramer's Rule*** to solve the following two simultaneous equations, determine x_1 and x_2:

$$2x_1 + 3x_2 = 7$$
$$4x_1 + 2x_2 = 18$$

X1=.....0.8..... X2=.....0.6667.....

(b) (6 Marks): Use the ***Gauss Elimination Method*** to solve the following three simultaneous equations. **Determine the values of X_1, X_2 and X_3**

$$2x_1 + x_2 + 3x_3 = 9$$
$$4x_1 + 3x_2 + 2x_3 = 16$$
$$-2x_1 + 2x_2 + x_3 = 3$$

X1=.....1.5625..... X2=.....2.5..... X3=.....1.125.....

6

a) $\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$ $\det A = (2.2) - (3.4) = -8$

$X_1 = \begin{bmatrix} 7 & 3 \\ 8 & 2 \end{bmatrix}$ $\det = (7.2) - (3.8) = -10$ $X_1 = \frac{-8}{-10} = 0.8$

$X_2 = \begin{bmatrix} 2 & 7 \\ 4 & 8 \end{bmatrix}$ $\det = (2.8) - (7.4) = -12$ $X_2 = \frac{-8}{-12} = 0.66667$

b) $\begin{bmatrix} 2 & 1 & 3 & 9 \\ 4 & 3 & 2 & 16 \\ -2 & 2 & 1 & 3 \end{bmatrix}$

$R_2 + -2R_1 = R_2$

$\begin{bmatrix} 2 & 1 & 3 & 9 \\ 0 & 1 & -4 & -2 \\ & & & \end{bmatrix}$

$R_3 = R_3 + R_1$

$\begin{bmatrix} 2 & 1 & 3 & 9 \\ 0 & -3 & -12 & -2 \\ 0 & 3 & 4 & 12 \end{bmatrix}$

$R_3 = R_3 + -3R_2$

$\begin{bmatrix} 2 & 1 & 3 & 9 \\ 0 & 1 & -4 & -2 \\ 0 & & & \end{bmatrix}$

Q3- Use the *Bairstow Method* applying first iteration, to find the roots of the polynomial: *(6 Marks)*

$F(x)=-2x^5+5x^4+17x^3-38x^2-36x+72$. Use (x^2+x-1), as a starting factor.

Given the results of the first synthetic division are:

$$b_5=-2, b_4=7, b_3=8, b_2=-39, b_1=11, b_0=22$$

P^*=..... q^*=..... x=.....